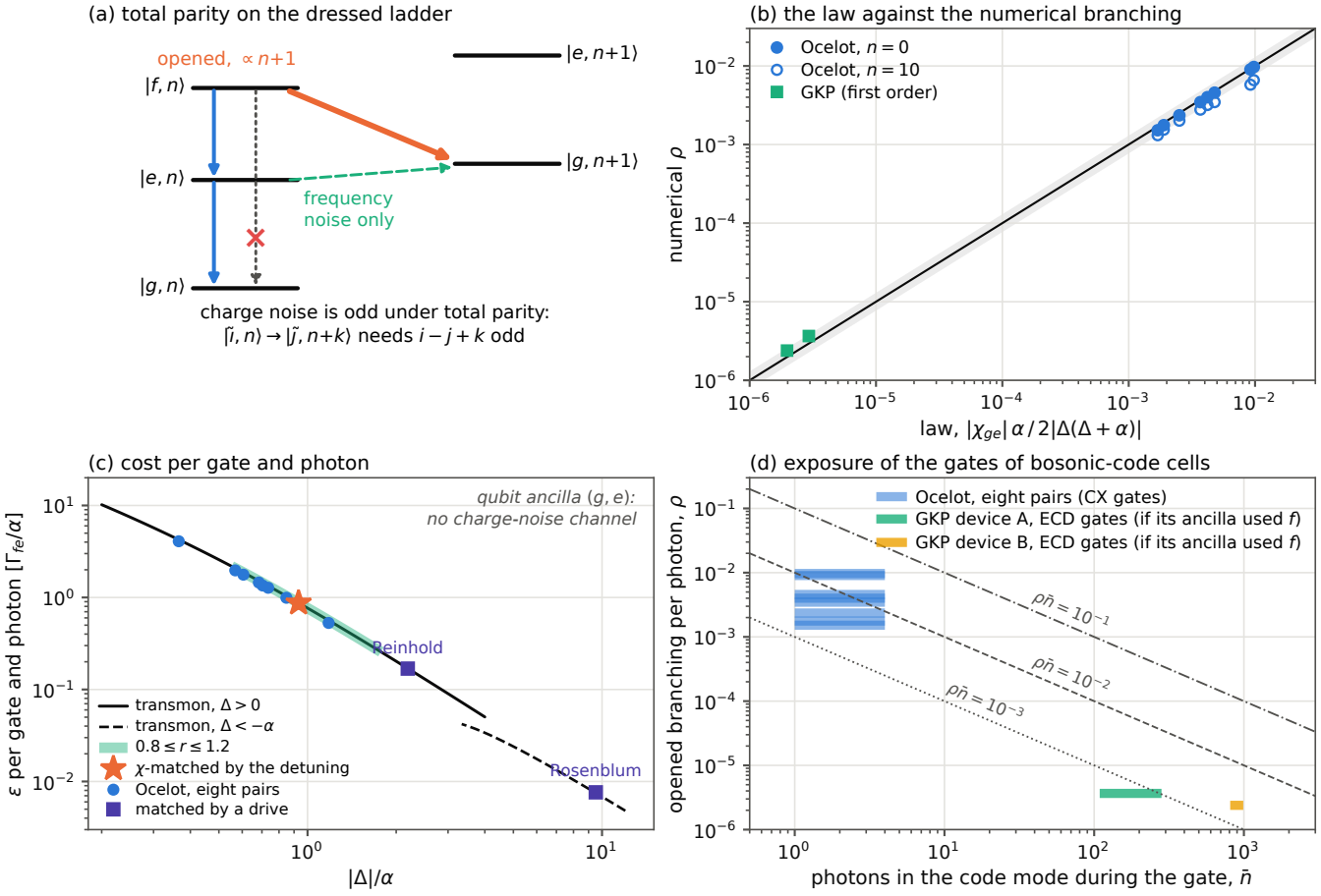
(a) total parity on the dressed ladder
opened, ∝ n+1
|f, n⟩
|e, n+1⟩
|g, n+1⟩
|e, n⟩
frequency
noise only
|g, n⟩
charge noise is odd under total parity:
⟨i, n⟩ → ⟨j, n+k⟩ needs i − j + k odd
(b) the law against the numerical branching
Ocelot, n = 0
Ocelot, n = 10
GKP (first order)
numerical ρ
law, |χge| α/2|Δ(Δ + α)|
(c) cost per gate and photon
qubit ancilla (g, e):
no charge-noise channel
Reinhold
Rosenblum
transmon, Δ > 0
transmon, Δ < −α
0.8 ≤ r ≤ 1.2
χ-matched by the detuning
Ocelot, eight pairs
matched by a drive
ε per gate and photon [Γφ/α]
|Δ|/α
(d) exposure of the gates of bosonic-code cells
Ocelot, eight pairs (CX gates)
GKP device A, ECD gates (if its ancilla used f)
GKP device B, ECD gates (if its ancilla used f)
ρn̄ = 10⁻¹
ρn̄ = 10⁻²
ρn̄ = 10⁻³
opened branching per photon, ρ
photons in the code mode during the gate, n̄